

Voice Spoofing Detection Based on the RVC Deep Learning CNN Model

Ban M. Alameri^{1, 2*}, *IEEE Member*

¹Department of Telecommunication
Engineering, E.T.S.I
University of Malaga
Malaga, Spain
²Electrical Engineering Department
Mustansiriyah University
Baghdad, Iraq

Jorge Munilla¹

¹ Department of Telecommunication
Engineering, E.T.S.I
University of Malaga
Malaga, Spain

Abstract—Advancements in artificial intelligence (AI) influence the security of voice-based authentication systems using techniques of deep learning. Deepfake voices, such as the Recursive Voice Conversion (RVC) model, have advanced significantly in recent years. This work presents an enhanced spoofing voice detection system that combines RVC used for generating deepfake voices as a pre-trained dataset, with MFCC spectrogram features and a convolutional neural network (CNN) model for training and classifying a spoofing dataset created using an RVC model based on real audio recordings to generate fake voices. The study aims to improve the accuracy and reliability of voice-based security systems against new threats from fake speech. The results show that the CNN model has the highest accuracy, at 99%, which contributes to securing biometric authentication and strengthening cybersecurity in many fields.

Keywords— Spoofing voices, Voice Authentication, CNN, MFCC spectrogram, RVC

I. INTRODUCTION

Speech is a significant norm of communication that is widely used in digital systems such as smartphones and applications. Further transmitting written content, speech conveys extra characteristics such as category, timing, and intonation, making it a valuable source of information. While the prospect of being misled by vocal imitations, particularly those of prominent figures, is disconcerting, the true severity of spurious or fraudulent audio recordings becomes evident within forensic settings [1]. Voice authentication leverages unique vocal characteristics to verify an individual's identity. Voice recognition technology enhances business operations by offering better security than traditional passwords and protecting sensitive information like financial details and health records [2].

The system of voice recognition technology has been significantly affected by the introduction of deepfake voice generators by AI, signifying a change shift in the domain of audio authentication [3]. This issue has been more critical with the advances in deep learning (DL), which have allowed the

creation of synthetic voices that closely mimic natural speech patterns, posing a major challenge to traditional voice verification. [4]. The deepfakes is based on variety of algorithms and techniques that seek to extract patterns and characteristics from noisy data. Many research examined a number of deep learning frameworks, including convolutional neural networks (CNN), assessing their effectiveness in the synthesis and detection of spoofing speech [5]. In addition, detecting such manipulated audio requires extracting features from speech signals to capture the different patterns of authenticity and deception [6]. With the advanced technology in the world, especially in artificial intelligence, it is necessary to enhancement the security of voice authentication against deepfake techniques , so the use of different models of machine learning has significantly impacted various areas of speech processing, including secure voice authentication [7].

The application of hybrid models of advanced neural network architectures (NNA) and sophisticated algorithms, researchers aspire to build accurate deepfake voice detection systems that are proficient in distinguishing authentic human voices from AI-generated replicas. This interdisciplinary methodology amalgamates the advantages of CNN and Long Short-Term Memory (LSTM) networks to facilitate real-time deepfake voice detection [8].

II. RELATED WORKS

In this section, we listed a related works of various studies has been undertaken to address the challenges posed by last approaches that suggest against artificial intelligence manipulation, including deepfake voice recognition.

A study by Mehrish, A. et al. looked at classical methods that use Gaussian mixture models (GMMs) and Mel-frequency cepstral coefficients (MFCCs) to pull out features. However, the limited adaptability of these methods to speech variations can result in higher false acceptance rates (FAR) [9]. V. Umarani et al. developed a hybrid MFCC for feature voice signals and employed a Bayesian GMM for model training. This hybrid model is effective at recognizing the

user based on voice and provides an accuracy of 94.5% for voice authentication[10] . Alireza Mohammadi et al. proposed a new approach to detect targeted data poisoning attacks in voice authentication. The study defines various attack scenarios against voice authentication systems. When the number of compromised samples is extremely low, the Librispeech dataset based on DNN models is particularly effective[11]. Nicolas M. Muller has used RawNet2. As one of the most popular models in voice anti-spoofing, RawNet2 directly processes the raw audio waveform and evaluates the performance of the trained models based on two metrics: overall test accuracy and individual F1 scores[12]. N.N. Prachi et al. have implemented a model utilizing deep learning approaches. CNN and LSTM are used in the TIMIT and LibriSpeech datasets, respectively.

The CNN model with the TIMIT dataset achieved 77.51% and 80.63% test accuracy for open set and closed set implementations, respectively[13].

K. Li et al. investigated how to protect the voice authentication system from data poisoning attacks and came up with a strong defense method called Guardian. It uses a convolutional neural network-based discriminator to improve current defenses. This approach was able to distinguish about 95% of attacked accounts from normal accounts[14].

The state-of-the-art study Mai et al. suggests that the underlying characteristics that distinguish bona fide speech from deepfakes are not yet fully understood, hampering the development of robust and generalizable detection systems[15].

Hamza and Others Emphasizes the use of machine learning algorithms and MFCC features in deepfakes for acoustic detection, examining different environmental conditions to improve the accuracy of the detection [16]. However, as the researchers note, the security of voice authentication systems remains a challenge, and understanding the precise features that humans use to identify deepfakes by leveraging the capabilities of deep learning to enhance voice authentication security could work towards ensuring the integrity and trustworthiness of voice authentication systems.

The current research is focusing on built a novel hybrid model based on RVC and CNN model to achieve more accurate than previous predecessors.

III. METHODOLOGY

In this section , a more logical and effective approach that uses an RVC model as a pre-processing step for a CNN-based voice recognition system. We propose to use Recursive Voice Conversion (RVC) as an effective data augmentation method. This combination could be particularly useful for improving the accuracy of speech recognition in challenging scenarios, such as when dealing with multiple speakers or noisy audio. Fig. 1 presents the study's basic proposal model framework.

A. Overview of the Retrieval-based Voice Conversion (RVC) Model

The Recursive Voice Conversion (RVC) model, developed recently by artificial intelligence researchers, offers a new approach to voice transformation. RVC is based on feature retrieval mechanisms using voice quality diffusion or neural networks (NN), while other voice mechanisms, such as neural vocoders, generative adversarial (GAN-based) models, etc., analyze sound waveforms directly. Unlike traditional voice conversion methods, which rely on imitation or modulation of voice characteristics, the RVC model uses a sophisticated retrieval mechanism to generate fake signals from the voice. The RVC works essentially by obtaining an existing voice sample from a database known as the source signal and applying advanced signal processing techniques to transform it into the desired target voice while retaining key characteristics such as tone, accent, and speech patterns. This ensures that the voice transferred retains its natural sound and is not indistinguishable from the real one.

Recursive Voice Conversion (RVC) works as a technique that is used to convert speech from one voice to another by using pre-recorded voice data by maintaining the natural characteristics of the original voice and analyzing and manipulating voice data. It retrieves segments from a reference voice and applies them to synthesized speech, making it sound more realistic [17].

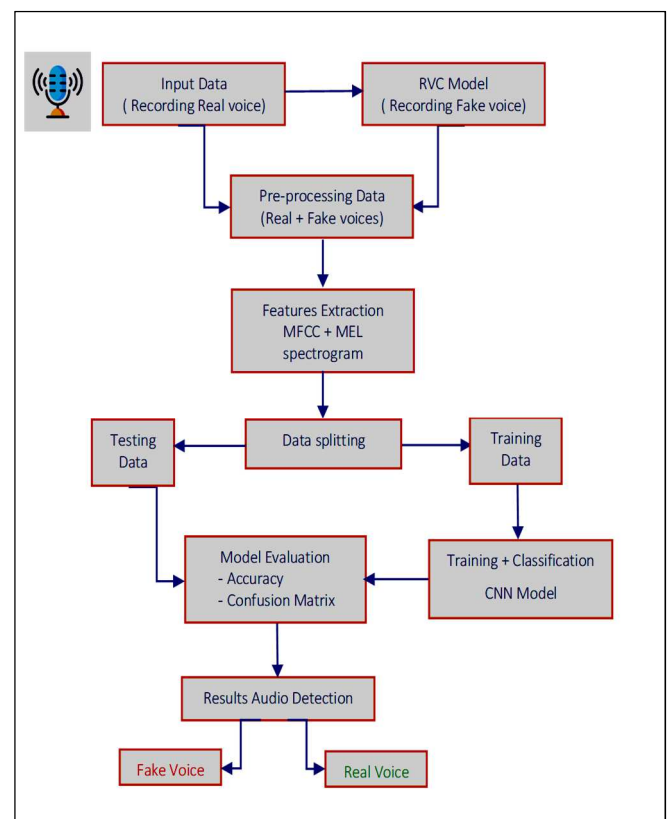


Fig. 1. Proposed model framework

B. Convolution Neural Networks (CNN)

CNNs have an effectiveness in processing audio signals that are represented as spectrograms. CNNs convert audio waveforms into visual representations of the spectrum of frequencies in a voice signal varying with time. CNN Layers are excellent for processing as feature detectors and use filters to scan the Mel-Spectrogram and identify patterns[18].

C. Features Extracted

Feature extraction is essential for detecting audio spoof attacks. This section provides a summary of the feature extraction, which is carried out utilizing Mel and MFCC spectrograms. These features capture the audio signal in a form that capable of being learned and be employed in the machine learning algorithms[19]. We use mel-spectrograms capture frequency-domain signals and train ensemble classifiers to distinguish genuine from deepfake speech.

- 1) Mel Spectrogram Extraction as show in Equation (1).

$$M(t, f) = 10 \log_{10} \left(\sum_{k=0}^{N-1} |X(t, k)|^2 \cdot H_{mel}(k, f) + \epsilon \right) \quad (1)$$

where:

- $X(t, k)$: is the STFT of input signal $x[n]$.
- $H_{mel}(k, f)$: is the Mel filter.
- ϵ : is a small constant ($1e-10$).

- 2) MFCC Computation as show in Equation (2)

$$MFCC_p(t) = \sum_{l=1}^L \log(M(t, l)) \cdot \cos\left(\frac{\pi p}{L} \left(l - \frac{1}{2}\right)\right) \quad (2)$$

Where for $p = 0, \dots, P - 1$ coefficients.

- CNN Model

- 1) Convolutional Layers as show in Equation (3).

$$C_{i,j,k}^{(l)} = \text{ReLU} \left(\sum_{m=0}^2 \sum_{n=0}^2 \sum_{c=0}^{F_l-1} W_{m,n,c,k}^{(l)} \cdot X_{i+m,j+n,c}^{(l-1)} + b_k^{(l)} \right) \quad (3)$$

where:

- $F_l \in \{32, 64\}$: are filters.
- Kernel size: is 3×3 .
- Stride: is 1, Padding is valid.

- 2) Max Pooling as show in Equation (4).

$$P_{i,j,k}^{(l)} = \max_{0 \leq m, n < 2} C_{2i+m, 2j+n, k}^{(l)} \quad (4)$$

D. Data set and Preprocessing

The data set in the experiment includes 17,870 voice samples, with an even distribution of 1,088 samples from deepfake and real in testing, 13,965 from fake and real in training, and 2,826 from fake and real audios in validation.

The size of data set is (1.4) GB .The dataset used for pre-training with real voices (recording voices) is available online on platform Hugging Face. A deepfake voices is generated by using RVC framework which based on a combination of

Hubert to achieve a realistic spoofing voice. As a result, we gathered a compilation of distinct voices from each individual. For this specific occasion, All the data, according to our research, is voice data that was divided into data that was 70% training, 15% validation, and 15% testing. Preprocessing is important because it improves the performance of machine learning algorithms by clean and well-structured data by eliminating and normalizes the undesirable elements from the data to prepare it for feature extraction and model training.

The input audio data is transformed into numerical data using three variables: pitch, resonance, and Mel spectrogram waveform, which represent the parameters of the system.

E. Model Implementation

The spoofing voice detection system is based on a scalable and modular architecture that ensures clarity and simplicity in experimentation[20].The system not only addresses current challenges in voice authentication but also sets a solid foundation for future advancements in security technology. The difference between the original and the converted voice is visually represented in the following figures to demonstrate the effectiveness of the RVC in producing a deepfake voice. The acoustic waveforms of the real and fake voices, mel spectrogram, and MFCC are shown in the following figures. Figure 2 shows a parallel comparison of the real and false voice waveforms, plotting the amplitude from -0.6 to 0.6 against a time interval of 0 to 2 seconds. Wave patterns share similar temporal patterns, confirming the preserved prosody of the major events of speech. However, the false signal shows slightly exaggerated amplitude peaks, e.g., close to 0.5s and 2s compared to the true signal, indicating a possible spectral enhancement during the conversion.

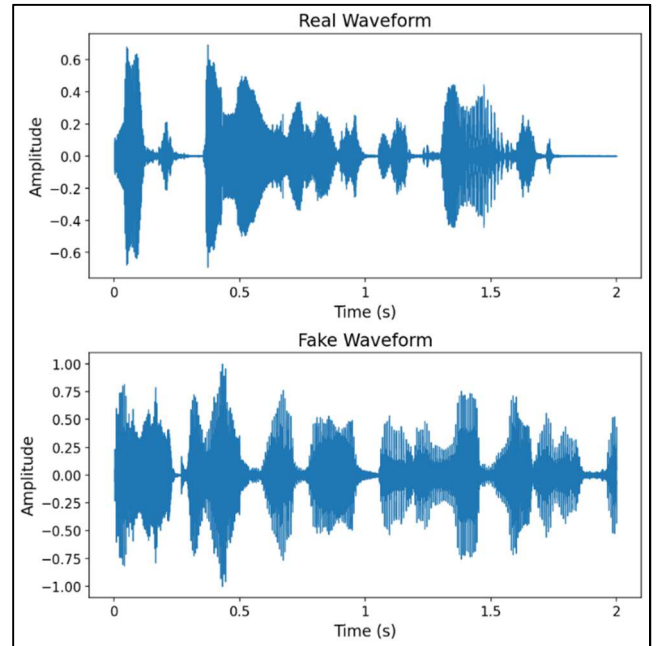


Fig. 2. Waveform Comparison of Real and fake Audio: visual difference between genuine recording and RVC deepfake audio

Figure 3 shows a comparative analysis of real and fake voice spectrograms. Both spectrogram results for real and fake share a similar temporal structure, with aligned features such as peaks near 0.5s and 2s, indicating preserved speech rhythm. Furthermore, the fake spectrogram exhibits distinct spectral differences, including enhanced high frequency evident by brighter hues above 256 Hz and a more uniform intensity distribution, suggesting fake spectral boosting or dynamic range compression. The real spectrogram, in contrast, shows greater dynamic variation, with more pronounced low-energy segments (darker regions). Frequency resolution values of 64 Hz to 512 Hz and a decibel scale of 0 dB to -80 dB further highlight these disparities.

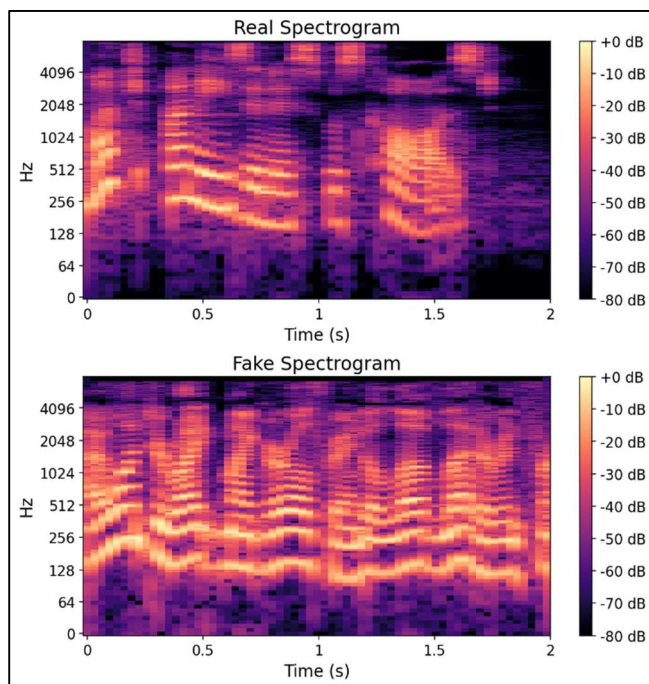


Fig. 3. Real and fake voices Spectrogram

These observations indicate that while the fake voice maintains temporal coherence, its spectral modifications, such as high-frequency emphasis and reduced dynamic range, may contribute to perceptible differences in clarity and naturalness compared to the real voice.

Figure 4 presents a parallel comparison of the real and false Mel spectrograms, which represent the voice signals over time (0-2 seconds) and frequency, ranging from 512 Hz to 4096 Hz. Both Mel-spectrograms use the frequency scale and display the intensity in decibels, ranging from -10 dB to -70 dB. The actual spectrum shows more natural energy fluctuations with clearly defined harmonic structures and gradual shifts in intensity between frequencies.

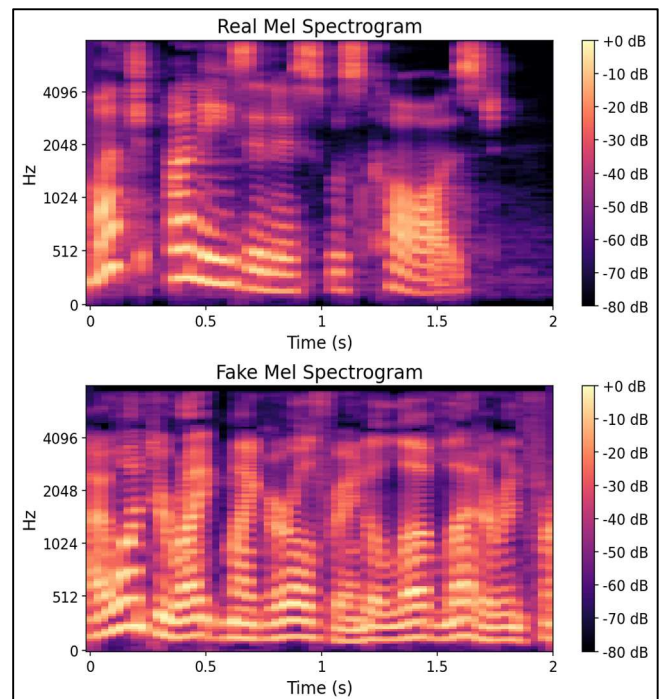


Fig. 4. Mel spectrogram of Real and Fake voices.

Figure 5 presents a comparison of real and fake Mel-Frequency Cepstral Coefficients (MFCCs), which are commonly used to represent the spectral characteristics of audio signals. The real MFCCs plot shows coefficient values ranging from approximately -400 to 200 over time from 0.5 to 1.5s, with a clear, smooth trajectory that reflects real speech dynamics.

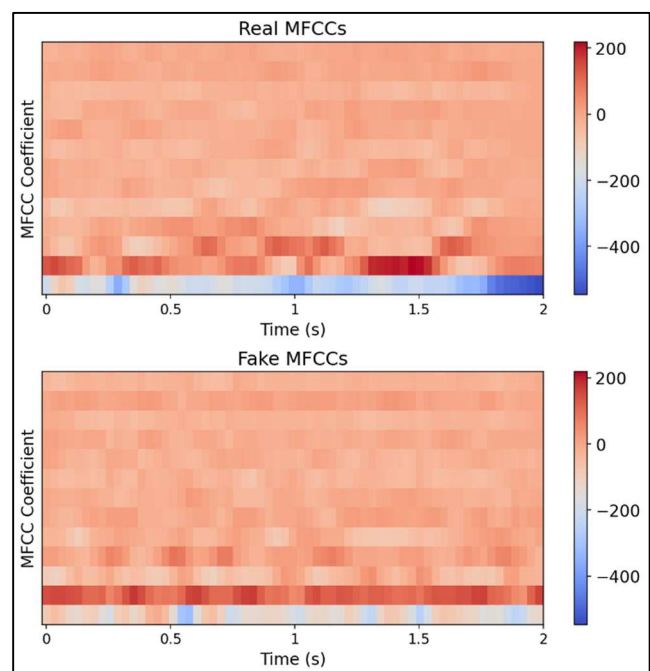


Fig. 5. MFCC of Real and Fake voices

In contrast, the fake MFCCs showed more erratic fluctuations, including abrupt spikes and dips near 0.5s and 1.5s, suggesting fake processing or synthesis artifacts.

On the other hand, both plots share some structural similarities; the fake version lacks the smooth transitions of the real voice, particularly in higher coefficient values. The difference indicates potential challenges in accurately reproducing the spectral analysis during voice modification, which may contribute to appreciable differences in audio quality. The maintained materialistic position peaks at approximately 1.5s, signifying successful inflection transfer, despite the deformed coefficient dynamic range, which reveals moderations in spectral fake voice.

IV. RESULTS AND METRIC

Figure 6 shows the training and validation performance metrics of the deep learning CNN model, showing both the accuracy and loss curves. The model demonstrated excellent learning capability, achieving a perfect training accuracy of 99% while maintaining a high validation accuracy of 97%. This packed alignment between the training and validation accuracies indicates sufficient generalization without overfitting. Likewise, the loss plot indicates a training loss of 0.04 and a validation loss of 0.08, with a minor gap, additionally reinforcing the model's robustness.

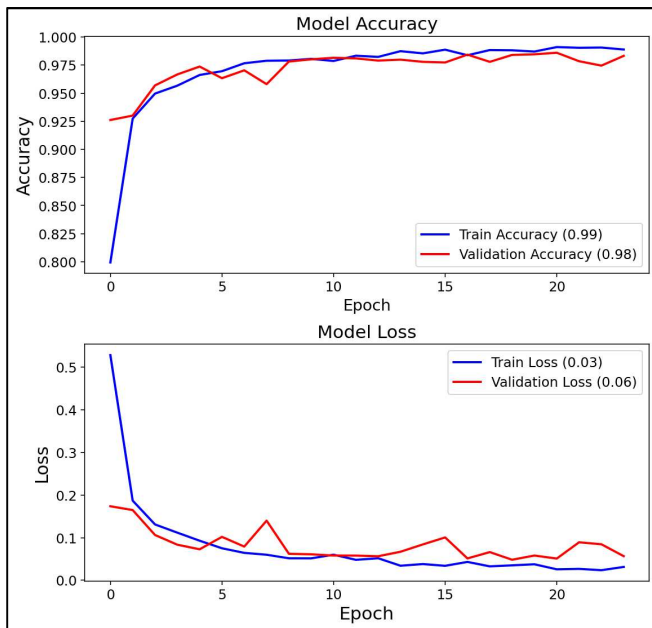


Fig. 6. CNN training and loss vs validation accuracy.

The consistently high performance across both metrics signifies that the model successfully learned meaningful patterns from the data while avoiding common pitfalls, such as overfitting or underfitting. These results reflect a well-trained model qualified to produce reliable predictions for both noticed and unnoticed data.

Figure 7 shows the Receiver Operating Characteristic (ROC) curves for the CNN model, which performed well in the observation. The ROC curve shows the ability to detect binary classifier training as its classification threshold increases. as

shown that CNN has an Area Under the Curve (AUC) of 1.00%. The modest boundary of CNN might be associated with its broader framework and its pre-training of the weights, which might additionally obtain deeper aspects of the data. This data is crucial for determining the models' efficacy in the setting of the investigation.

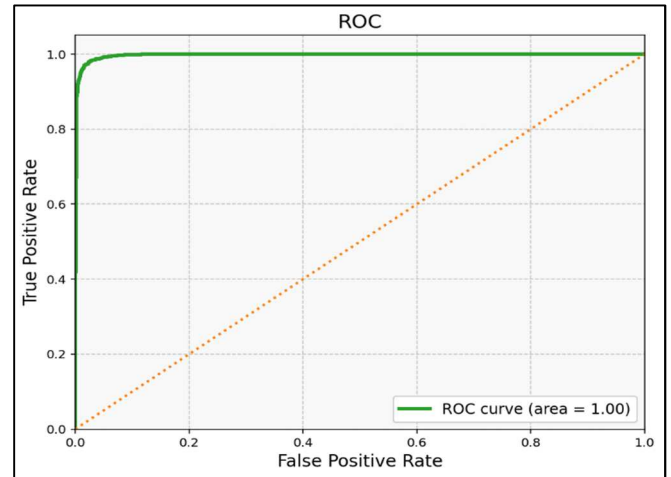


Fig. 7. ROC curve for CNN

Figure 8 shows a matrix of confusion, indicating how well the spoofing voice detection algorithm can distinguish between real and deepfake voices. The matrix has four horns, as shown in figure 8 below. In the upper-left corner are the 1,728 real positives, which are precisely labeled as real voices. There were 48 fake negatives and fake voices in the right upper quadrant. Satisfied that the real votes in the lower left quadrant had been incorrectly classified as 34 false positives. The 1,764 deepfake voices were correctly identified as the true negative in the lower right quadrant.

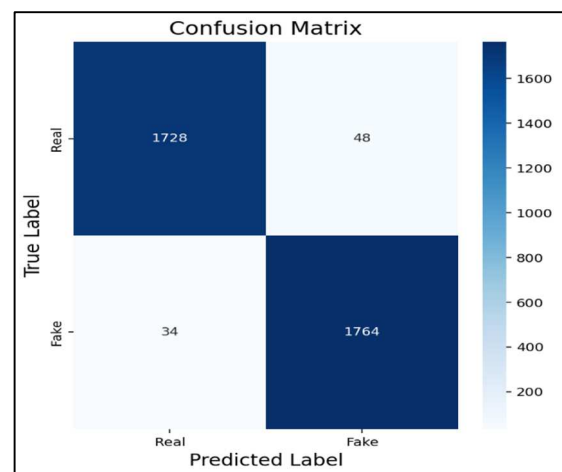


Fig. 8. CNN Confusion Matrix

The model successfully represents both the real and the false voices according to this matrix. The model was reliable, although only 48 fake voice misclassifications indicated a slight difference.

V. CONCLUSION

This article is focusing on voice authentication security by building a model for detecting a spoofing voice. This approach is based on generating deepfake audio by using the system for pre-trained data and combining it with a CNN model for classification. The CNN model used the database, which was developed based on the RVC model, and analyzed a dataset of 17,870 real and fake voice samples. The CNN model had the highest accuracy, at 99%. In addition, validation accuracy did not show any overfitting, and loss graph models performed well on training data. CNN models generalize the data that is necessary for effective voice verification. The deep learning technique enhances the workflow, encouraging seamless detection of fake voices. Future research could be improved by using cross-domain transfer learning to create hybrid models integrating CNN and SLTM Long Short-Term Memory trained on a large and diverse set of voice data.

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DATA AVAILABILITY STATEMENT

The dataset used is available upon reasonable request.

CONFLICTS OF INTEREST

The authors declare no conflicts of interest

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